Doubling Every ~7 Months

Measuring AI Ability to Complete Long Software Tasks

Time Horizon Doubling Every 7 Months

METR Research Team • arXiv:2503.14499 • March 2025

A New Metric: 50% Task-Completion Time Horizon

1

Assemble Task Suite

170 diverse software/research tasks from 3 seconds to 8 hours

→

2

Measure Performance

Time skilled humans + measure AI agent success rates

→

3

Fit Logistic Model

Find task duration where AI succeeds 50% of the time

```
50% Time Horizon = Task duration at which P(success) = 0.5
```

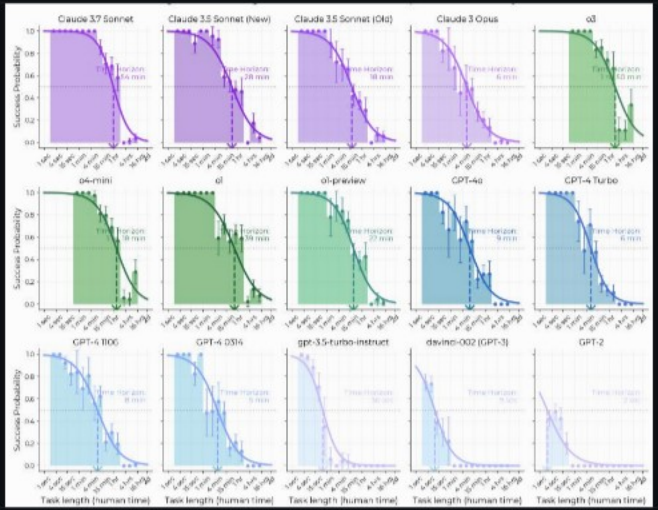
Family	Length	Description
find_shell_script	3 seconds	Multiple choice: "Which file is a shell script?" Choices: "run.sh", "run.txt", "run.py", "run.md"
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

170 Tasks Spanning 3 Seconds to 8 Hours

97	7	66
HCAST 1 min — 30 hrs	**RE-Bench** All 8 hours	**SWAA** 1 sec — 30 sec

Task	Duration	Description
find_shell_script	3 seconds	Multiple choice: "Which file is a shell script?"
wikipedia_research	1 minute	Research simple factual information from Wikipedia
oxdna_simple	9 minutes	Detect and fix a bug in molecular dynamics simulation input files
munge_data	56 minutes	Write a Python script to transform JSON data by inferring conversion rules
cuda_backtesting	8 hours	Implement custom CUDA kernels for 30x speedup of a backtesting tool

AI Task Horizon: 2 Seconds (GPT-2) to ~2 Hours (o3) in 6 Years



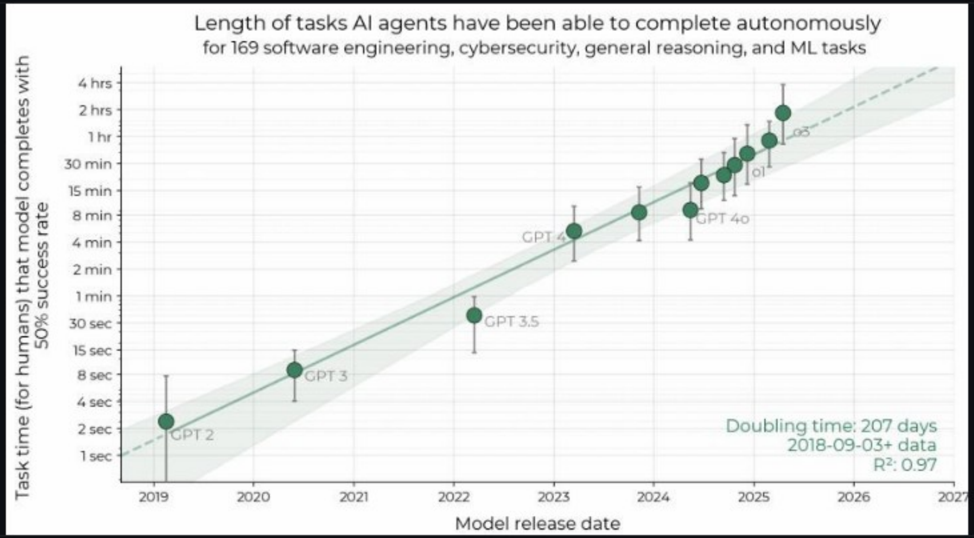
~207 days
DOUBLING TIME

R^2 = 0.97
FIT QUALITY

~2,000x
GROWTH (2019—2025)

GPT-2	GPT-4 (0314)	Claude 3.7 Sonnet	o3
~2 sec	~5 min	~54 min	~2 hrs

Success Probability Drops Sharply as Task Length Increases



Key Observations

- → All models show steep decline in success rate as task duration increases
- → Frontier models (Claude 3.7 Sonnet, o3) push the curve significantly rightward
- → GPT-2: 50% horizon at ~2 seconds
- → o3: 50% horizon at ~2 hours (3,600× longer than GPT-2)
- → 80% time horizon is ~5× shorter than 50% horizon across all models

Reasoning, Tool Use, and Error Recovery Drive Time Horizon Growth

🧠 **Improved Logical Reasoning**
Deeper chain-of-thought processing enables longer, more complex task chains

🔧 **Better Tool Use**
Enhanced ability to invoke APIs, execute code, browse the web, and manipulate files

🔄 **Error Recovery**
Greater self-awareness and ability to detect, diagnose, and recover from mistakes

✨ **Reliability & Adaptation**
More consistent execution and better adaptation to unexpected situations

Limitation: Performance is notably lower on less structured, "messier" tasks. External validity remains an open question — tasks may not perfectly represent the average segment of real-world intellectual labor.

Extrapolation Predicts Month-Long Task Automation by 2028–2031

2028—2031

Projected 50% time horizon reaches 1 month (167 work hours)

Month-Long Task Automation

2025 (o3: ~2 hrs) 2031

2025: Current frontier at ~2 hours (o3)

2026—2027: Estimated 8—16 hour horizon

2028—2031: 1-month (167 hr) threshold reached

Note: 2023—2025 growth rate ~20% faster than 2019—2025 average

Caveats: Trend may slow due to saturation effects, capability cliffs, or external factors. External validity of task suite to real-world job distributions remains uncertain.

Takeaways: Exponential Autonomous Capability Growth Demands Proactive Safety Governance

1. **Robust Metric:** 50% time horizon is a robust, intuitive metric for tracking AI autonomy with $R^2 = 0.97$ over 6 years of model releases.

2. **Exponential Growth:** Doubling every ~7 months with possible acceleration since 2024. Current growth rate 20% faster than historical average.

3. **Current Frontier:** o3 can autonomously complete tasks taking skilled humans nearly 2 hours — up from 2 seconds for GPT-2 in 2019.

4. **Open Questions:** External validity remains uncertain — tasks may not perfectly represent real-world intellectual labor distributions.

5. **Key Limitations:** Poor performance on unstructured/"messier" tasks. Trend extrapolation may not hold indefinitely.